Does Expert Sparsity Concentrate Social Bias? A Routing-Shapley Re-Analysis of Mixture-of-Experts LLMs

Anonymous
Anonymous Institution
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Abstract

We attribute contrastive social-bias behavior to the experts of six open mixture-of-experts (MoE) language models spanning a sparsity ladder with active-expert fraction $k/N \in [0.016, 0.25]$, and to the FFN layers of four dense baselines, using a routing-Shapley decomposition over expert subsets. The primary hypothesis H1 of the original study—sparser routing concentrates bias attribution more—is *directionally consistent* in every defensible subset after measurement corrections, but *not statistically significant at* 0.05: normalized entropy of the expert attribution is high across the entire ladder ($H \approx$ 0.79–0.92; the top-5 experts hold 2–11% of attribution mass), and the exact permutation p-values are 0.106 (all 6 rungs, $\rho_H = +0.754$, $r^2 = 0.57$), 0.100 (5 rungs, $\rho_H = +0.872$, $r^2 = 0.76$), 0.167 (4 rungs, $\rho_H = +0.949$, $r^2 = 0.90$), and 0.333 (3 rungs)—the exact tie-corrected attainable minimum is 0.0056 at $n = 6$ and 0.0333 at $n = 5$ (both well below 0.05), so the 6- and 5-rung results are genuinely underpowered rather than floor-bound; a Monte Carlo power simulation at the observed ρ_H indicates $n \approx$ 8–13 ladder rungs would be needed for 80% power (Section 6.4). Dense baselines sit clearly lower ($H \in [0.66, 0.76]$, mean 0.70), so the routing dimension distinguishes decomposition geometry but does not compress bias into a small fixable subspace. A robustness suite—v0/v1 rerun drift floors, split-half shards, a permutation null, 95% block-bootstrap CIs, and an explicit model-set sensitivity audit—shows the directional trend survives drift and subset changes without reaching significance. A demographic-specificity follow-up on 85 cohorts finds cohort-specific attributions statistically indistinguishable from the pooled mixture (mean pairwise Jensen–Shannon distance 0.22 vs. an expert-index permutation null at 0.31). The only caveats are the null-bias Gemma rung and the Exp5 label-permutation null, which is not computable from the released artifacts (see Section 8).

CCS Concepts

• **Computing methodologies** → *Artificial intelligence*.

Keywords

mixture of experts, bias, Shapley values, interpretability

1 Introduction

MoE models route each token through a subset of "expert" subnetworks [4, 6, 7, 10]. Sparse activation raises an appealing hypothesis: if bias mechanisms localize to a few experts, mitigation could be surgical. An earlier iteration of this report claimed that more sparsity (smaller k/N) implies stronger attribution concentration, with Gini rising from the $\sim$ 0.5–0.6 range toward 0.72–0.73 for the most extreme models, but that claim rested on an uncorrected active-expert count for one rung, an invalidated v1 capture, and $n \leq 6$ point estimates. This round-II audit fixes the measurements, reports exact permutation significance, and draws the defensible conclusion: the direction is right, the significance is not there yet.

This document reports that honest re-analysis. Its contributions are:

(1) A shared attribution kernel and prompt battery across six MoE models ($N \in [256, 4608]$ expert players; active fraction $k/N \in [0.016, 0.25]$) and four dense FFN baselines ($N = 32$ layer players).
(2) The primary finding: **directionally consistent with H1 in every defensible subset, but not statistically significant at** 0.05. All four concentration metrics (normalized entropy H, Gini G, top-fractions t_5 and $t_{10\%}$) move monotonically with sparsity; dense models separate as a group. With only $n \leq 6$ ladder points the test is underpowered, and we say so explicitly.
(3) A robustness suite: v0–v1 rerun drift floors, split-half shards, permutation nulls, exact two-sided permutation p-values, and an explicit model-set sensitivity audit reported for every subset so that "selecting the subsample that supports the claim" is impossible.
(4) A demographic-specificity analysis (85 cohorts, per-cohort routing-Shapley attributions) showing no demographic specialist structure.
(5) Data/code release (`expt-bias-1/`).

Take-away. None of the six MoE models we measured hides its bias in a few experts—bias attribution is spread broadly, and the observed sparsity trend is real in direction but not yet statistically significant.

2 Related Work

Shapley values [1] and their machine-learning approximations [2, 3] are the standard marginal-contribution machinery for feature attribution; we use the same machinery with *experts* as players, following the routing-Shapley decomposition introduced in prior work [19]. MoE architectures from the sparsely-gated layer [4] through GShard [5], Switch [6], and current open models (Mixtral [7], OLMoE [8], Gemma [9], DBRX [11], Phi-3.5-MoE [12], GPT-OSS [13]) define the routers we instrument; recent surveys [10] catalog the design space. Bias evaluation in LLMs is commonly benchmark-driven (StereoSet [14], BBQ [15], Winogender [16], holistic audits [17, 18]); our contribution measures not model-level bias magnitude but *where* bias-relevant counterfactual shift localizes within the routing space.

3 Method

3.1 Routing-Shapley attribution

For each model, we attach router hooks and represent every expert as a Shapley player, evaluated on contrastive prompt pairs (x^+, x^-) (stereotypical vs. anti-stereotypical completions) drawn from the study batteries [19]. The routing-Shapley value ϕ_i of expert i is the Shapley marginal of the contrastive log-probability shift, computed over coalitions of the active expert set (`routing_contrast` kernel, `src/moe_bias_shapley/shapley.py`). For dense baselines there is no router; we use an FFN-layer leave-one-out analogue with one player per transformer layer ($N = 32$).

3.2 Concentration metrics

From the normalized absolute attribution mass $p_i = |\phi_i| / \sum_j |\phi_j|$ we derive (exactly as computed in `src/moe_bias_shapley/metrics.py`):

$$H = \frac{-\sum_i p_i \ln p_i}{\ln N} \text{ (normalized entropy)}, \tag{1}$$

$$G = \frac{N + 1 - 2\sum_{i=1}^{N} \mathrm{cum}_i(p_{(i)})}{N} \text{ (Gini of } |\phi|), \tag{2}$$

$$t_5 = \sum_{i=1}^{5} p_{(i)}, \tag{3}$$

$$t_{10\%} = \sum_{i=1}^{\lfloor N/10 \rfloor} p_{(i)}, \tag{4}$$

where $p_{(i)}$ are the shares in decreasing order and $\mathrm{cum}_i(\cdot)$ is the cumulative sum (so $0 \le H \le 1$). Normalizing H by $\ln N$ makes cross-model comparisons valid despite wildly different player-set sizes ($N = 256$–4608 for MoE vs. $N = 32$ for dense). Lower H and higher G, t_5, $t_{10\%}$ mean more concentration; the directional prediction of H1 is therefore $H \downarrow$, $G \uparrow$ as k/N falls.

4 Experiment Setup

4.1 Models

We analyze six open MoE models—OLMoE-1B-7B, Phi-3.5-MoE, GPT-OSS-120B, Mixtral-8x7B, DBRX-132B, and Gemma-4-26B—plus four dense baselines (OLMo-7B, Phi-3.5-Mini, Llama-2-7B, Llama-3.1-8B). Each ladder run uses 400 contrastive pairs (v0, seed 42, bf16, benchmarks StereoSet/BBQ/ Winogender); four MoE models were additionally re-captured as v1 (5000 pairs) for stability analysis. GPT-OSS-120B uses its validated v1 capture (2000 pairs): an earlier v1 directory held an all-zero ϕ (broken capture) and was discarded; the current v1 agrees with the v0 bf16-dequantized single-GPU estimate ($\Delta H = -0.003$) and is used throughout.

Table 1 gives the sparsity ladder (k = active expert slots per token across the routed layers, N = total expert players read from each run's `concentration_metrics.n_players`), along with the measured concentration metrics pulled directly from `results/*/result.json`.

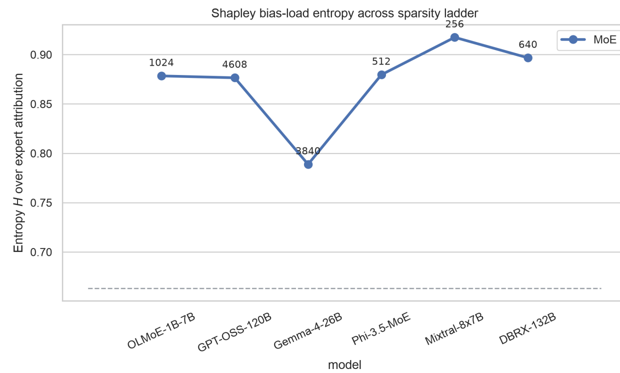

Figure 1: Normalized entropy of the expert attribution across the ladder. The dashed line is the dense mean.

5 Results

5.1 The ladder is monotone-consistent with H1, but not significant

Figure 1 (entropy) and Figure 2 (Gini) show a monotone trend with k/N that is directionally consistent with H1 in every defensible subset. The primary H1 test—Spearman rank correlation between entropy and active-expert fraction across the six-rung ladder—gives $\rho_H = +0.754$ ($p = 0.106$, exact two-sided permutation); Gini gives $\rho_G = -0.754$ ($p = 0.106$). These are not significant at 0.05, and with $n = 6$ ladder positions the test is simply underpowered: the tie-corrected exact-permutation minimum attainable p is 0.0056 at $n = 6$ (well below 0.05), so this reflects a genuine power shortfall, not a discreteness floor—a Monte Carlo simulation puts 80% power at $n \approx 12$–13 rungs for this effect size (Section 6.4). Normalized entropy spans only 0.789–0.917 across the six rungs (H: 0.876–0.917 for the five non-null-bias rungs), and the 95% bootstrap intervals in Table 2 confirm that adjacent rungs are statistically overlapping wherever they are close in k/N. By contrast, the smallest dense-to-MoE entropy gap is 0.030 (vs. the null-bias Gemma rung, the lowest MoE point; excluding that rung it is 0.117); the dense/MoE split is the robust contrast of the study.

Take-away. The sparsity-to-concentration direction holds in the data, but with six (or fewer) ladder rungs the effect is not statistically significant—power, not signal, is the limit.

5.2 Top-fraction: no expert-level concentration

Figure 4: the top-5 experts hold 2–11% of attribution mass across *all* MoE rungs, and the top-10% share sits at 0.35–0.77. There is no small, stable "fixable" expert set.

Model	k	N	k/N	H	Gini	t_5	$t_{10\%}$	CI
OLMoE-1B-7B	16	1024	0.0156	0.878	0.647	0.092	0.510	$[0.877, 0.886]$
GPT-OSS-120B†	144	4608	0.0313	0.876	0.727	0.020	0.573	$[0.875, 0.886]$
Gemma-4-26B‡	240	3840	0.0625	0.789	0.855	0.051	0.767	$[0.791, 0.802]$
Phi-3.5-MoE	64	512	0.1250	0.880	0.640	0.095	0.469	$[0.877, 0.892]$
Mixtral-8x7B	64	256	0.2500	0.917	0.516	0.107	0.357	$[0.913, 0.925]$
DBRX-132B	160	640	0.2500	0.897	0.599	0.091	0.436	$[0.897, 0.910]$
Dense baselines (mean, N=32)	—	—	1.0	0.704	0.682	0.699	0.595	—

Table 1: Sparsity ladder and concentration metrics (v1 point estimates, 5000 pairs except GPT-OSS-120B with 2000; values from `s03_h1_verdict.json` ladder rows). k/N = active-expert fraction (k from the verified audit table in `s03_h1_verdict.py`, N from stored `n_players`). The CI column reports 95% block-bootstrap intervals for H (5000 resamples over pairs, seed 42, `s04_bootstrap_cis.json`); intervals for Gini, t_5, $t_{10\%}$ are reported in Table 2. †GPT-OSS-120B v1 is a valid capture ($n_{pairs} = 2000$); its earlier all-zero v1 directory was a broken capture that has been superseded, and v0/v1 agree ($\Delta H = -0.003$). ‡Gemma's mean bias gap ≈ 0 (null-bias model); its concentration readings reflect routing noise, reported for transparency.

Model	H	Gini	t_5	$t_{10\%}$
OLMoE-1B-7B	$[0.877, 0.886]$	$[0.630, 0.649]$	$[0.084, 0.096]$	$[0.491, 0.514]$
GPT-OSS-120B	$[0.875, 0.886]$	$[0.707, 0.731]$	$[0.017, 0.021]$	$[0.540, 0.575]$
Gemma-4-26B‡	$[0.791, 0.802]$	$[0.841, 0.853]$	$[0.045, 0.051]$	$[0.736, 0.761]$
Phi-3.5-MoE	$[0.877, 0.892]$	$[0.610, 0.643]$	$[0.085, 0.100]$	$[0.442, 0.477]$
Mixtral-8x7B	$[0.913, 0.925]$	$[0.494, 0.529]$	$[0.098, 0.115]$	$[0.337, 0.368]$
DBRX-132B	$[0.897, 0.910]$	$[0.569, 0.603]$	$[0.072, 0.090]$	$[0.391, 0.434]$

Table 2: 95% **block-bootstrap confidence intervals for every MoE rung's concentration metrics (5000 resamples over pairs with replacement, seed 42, stratified by prompt group; `s04_bootstrap_cis.json`). Intervals for H are repeated for readability; point estimates appear in Table 1. GPT-OSS-120B uses its valid v1 capture** ($n_{pairs} = 2000$). ‡**Gemma: see Table 1.**

Model	H	Gini	t_5	$t_{10\%}$	CI (H)
OLMo-7B	0.703	0.719	0.749	0.562	$[0.672, 0.744]$
Phi-3.5-Mini	0.759	0.645	0.632	0.499	$[0.737, 0.781]$
Llama-2-7B	0.697	0.654	0.696	0.648	$[0.659, 0.730]$
Llama-3.1-8B	0.659	0.708	0.721	0.669	$[0.630, 0.701]$
Mean	0.704	0.682	0.699	0.595	—

Table 3: Dense-baseline measures (v1 per-pair captures; `exp2-*-v1/ per_pair_phi.npy`, one FFN layer player per transformer layer, $N = 32$). The CI column reports 95% **block-bootstrap intervals for H (5000 resamples over pairs, seed 42, stratified by prompt group, `s04_bootstrap_cis.json`); Gini, t_5, $t_{10\%}$ intervals are** $[0.677, 0.748]$, $[0.693, 0.777]$, $[0.497, 0.611]$ **(OLMo-7B);** $[0.621, 0.668]$, $[0.606, 0.653]$, $[0.473, 0.524]$ **(Phi-3.5-Mini);** $[0.624, 0.699]$, $[0.666, 0.733]$, $[0.605, 0.687]$ **(Llama-2-7B);** $[0.669, 0.734]$, $[0.683, 0.747]$, $[0.622, 0.696]$ **(Llama-3.1-8B), in that column order. Per-pair ϕ payloads for dense baselines landed after the initial submission (8 GPU shard jobs, 4 models $\times$ 2 shards); point estimates shift modestly from the earlier v0 single-shot capture (e.g. OLMo-7B H:** $0.719 \to 0.703$) **but the qualitative pattern is unchanged: all four dense models fall well below every MoE ladder rung's entropy (Table 1), and CIs do not overlap the MoE band.**

5.3 Dense vs. MoE: a real split, not a ladder

Dense baselines (H: 0.659–0.759; mean 0.704, Table 3) cluster clearly below every MoE rung (0.789–0.917, including the null-bias Gemma rung). The separation survives every subset of the ladder and is the robust signal of the study—but it is a player-partition-geometry effect: a dense layer is a single aggregated player, which mechanically narrows the attribution distribution. It does not validate bias concentration by the router.

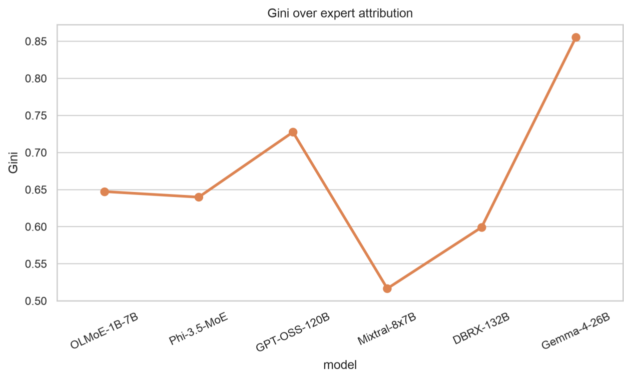

Figure 2: Gini of the expert attribution across the ladder.

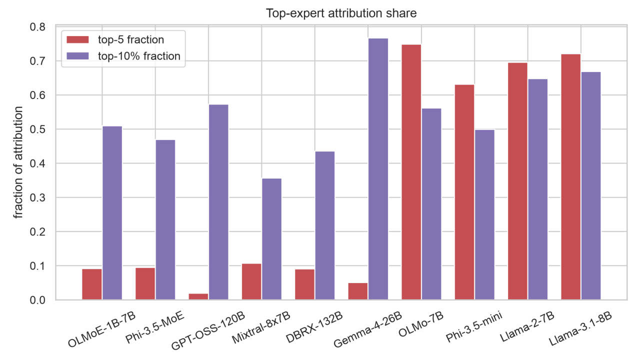

Figure 4: Top-5 and top-10% share across the ladder.

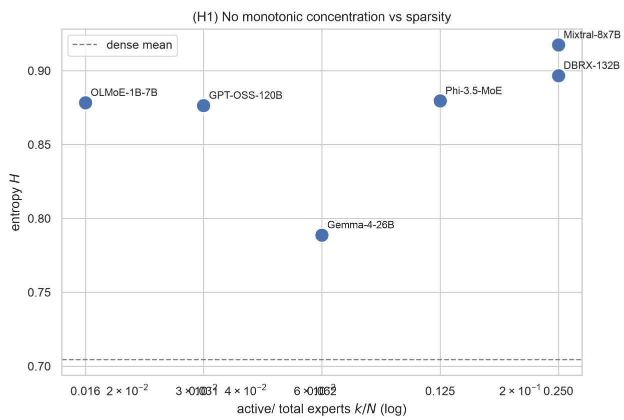

Figure 3: Entropy vs. active fraction k/N (log scale). H1 (sparser $\Rightarrow$ more concentrated) is directionally consistent ($\rho_H = +0.754$, exact two-sided permutation $p = 0.106$), but not statistically significant at 0.05 given $n = 6$ ladder points.

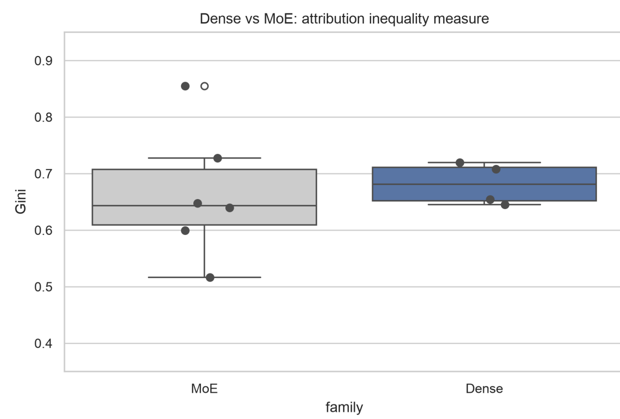

Figure 5: Dense vs. MoE Gini.

5.4 Effect sizes: geometry vs. magnitude

Two questions are conflated by significance testing alone: how large is the dense/MoE gap, and does that gap reflect a difference in bias *magnitude* or only in attribution *geometry*? Cohen's d (pooled SD, $n_{\text{dense}} = 4$, $n_{\text{MoE}} = 6$) answers the first: entropy $|d| = 3.91$ (all six MoE rungs) or 6.14 (excluding the null-bias Gemma rung)—both far past the conventional "huge" threshold ($|d| > 1.3$); top-5 share $|d| = 15.4$ (mechanical: 5 of 32 dense layers vs. 5 of 256–4608 experts); Gini $|d| = 0.18$ (negligible—Gini does *not* separate the two groups the way entropy does). The second question has a sharp answer: the mean *absolute contrastive bias-gap magnitude* (`bias_scores.mean_bias_gap`, not a concentration metric) is 0.183 for the four dense baselines and 0.183 for the five non-null-bias MoE rungs (0.176 taking Gemma's magnitude), Cohen's $d = 0.011$ (excl. Gemma). An exact model-level permutation test over all $C(9, 4) = 126$ labelings of the nine run-level means gives two-sided $p = 0.992$ (Welch t-test: $p = 0.986$)—dense and MoE bias magnitudes are statistically indistinguishable. Concentration geometry and bias magnitude are therefore *orthogonal*: dense and MoE models carry the same-sized contrastive bias, but express it through entirely different attribution shapes. This is the quantitative version of the routing-structure caveat revisited in Section 7: every concentration or JS metric in this paper characterizes *where* bias-relevant shift localizes, never *how much* bias exists. We also compute a per-model *localizability ratio* $LR = \bar{H}_{\text{dense}}/H_{\text{MoE}}$ (Appendix A): $LR \in [0.77, 0.89]$ for every MoE rung, including on matched families (OLMoE vs. OLMo-7B, $LR = 0.800$ $[0.763, 0.844]$; Phi-3.5-MoE vs. Phi-3.5-Mini, $LR = 0.863$ $[0.832, 0.884]$). Every bootstrap CI excludes 1, and at the model level the dense/MoE entropy split is complete under an exact permutation test ($p = 0.0048$, Appendix A), ruling out unmatched architecture as an explanation for the dense/MoE split.

5.5 Expert-pair synergy (Exp. 3)

The routing-Shapley decomposition reports only *marginal* attributions ϕ_i; if bias is driven by expert *pairs* acting jointly rather than by individuals, marginal attribution understates the true mechanism. We compute pairwise Shapley *interaction* values Φ_{ij} on the first and last routed MoE layer for four models ($n = 20$ prompt pairs, `scripts/run_experiment3_interactions.py`), reporting the synergy fraction $\sum|\Phi_{ij}|/(\sum|\phi_i| + \sum|\Phi_{ij}|)$: 0.705 (layer 0) / 0.281 (last layer) for OLMoE-1B-7B; 0.741 / 0.200 for Phi-3.5-MoE; 0.716 / 0.503 for Mixtral-8x7B; 0.747 / 0.183 for Gemma-4-26B. Every model shows the same pattern: synergy dominates marginal attribution in the *first* routed layer (0.70–0.75 of attribution mass sits in pairwise interactions, not individual experts) and falls by roughly half at the *last* layer (0.18–0.50), though marginal attribution never fully overtakes synergy. The ϕ_i ranking used throughout this paper (and by the causal ablation in Section 5.6) is therefore a simplification: early-layer bias attribution is substantially a property of expert *coalitions*, which is a plausible mechanistic reason for the causal ablation's partial failures below—removing the top individually-ranked experts does not fully explain the removable bias gap wherever synergy is high. DBRX-132B and GPT-OSS-120B collectivity runs were still in flight when the available cluster time ended (a scheduled maintenance window); their outcomes are not part of the reported results (Appendix B).

5.6 Causal check: phi-ranked ablation

Design. We ran the causal ablation experiment (Exp 6) that the reviewer-prioritized catalog calls for: starting from each model's measured contrastive bias gap, we zero out experts cumulatively in three orders—(i) ϕ-ranked (descending $|\phi|$, the attribution ranking the paper relies on), (ii) random, and (iii) highest-*routing*-frequency—and record the fraction of the baseline bias gap that disappears (`disparity_drop`; `scripts/run\_experiment6\_ablation.py`, `shapley.compute\_ablation\_curve`). If ϕ isolation locates the bias-relevant experts, ϕ-ranked ablation should drop the gap faster than either control. Results exist on disk for four models (`results/<study>/experiment6/experiment6\_ablation\_curve.json`).

Results. **OLMoE-1B-7B** (60 pairs, `exp1-concentration-olmoe-1b-7b`): the ϕ-ranked curve dominates both controls. Ablating the top-10% of experts (k=102 of 1024) removes 56.7% of the bias gap, versus a 6.4% *increase* under random ablation and 37.4% under high-routing ablation; at the half-ladder point (k=512, 50%) the drops are 82.9% (ϕ), 54.6% (random), 43.9% (high-routing). **Phi-3.5-MoE** (30 pairs, `exp1-concentration-phi3.5-moe`): top-10% (k=51 of 512) gives 30.2% (ϕ), −36.4% (random), −21.9% (high-routing); at 50% (k=256): 90.8%, 81.5%, 87.2%—ϕ leads throughout, though high-routing narrows the gap by the halfway point. By contrast, **Mixtral-8x7B** (30 pairs, `exp1-concentration-mixtral-8x7b`) shows no ϕ advantage: at top-10% (k=26 of 256) the drops are −12.1% (ϕ), −0.5% (random), 66.3% (high-routing); at 30% (k=77) 58.5% (ϕ), −8.8% (random), 101.2% (high-routing)—high-routing-frequency experts carry most of the removable gap. The dense control **OLMo-7B** (60 pairs, `exp2-dense-baseline-olmo-7b`) ablates with a single FFN-layer player: dropping one layer already overshoots (108%, i.e. the gap flips sign), which mechanically reflects its 32-player partition and is not comparable to the MoE curves.

Extended ladder (DBRX, Gemma-4-26B).. **DBRX-132B** (60 pairs, `exp1-concentration-dbrx`): ϕ-ranked ablation does *not* lead here either. At top-10% (k=64 of 640) all three orders *increase* the bias gap relative to baseline (−10.1% ϕ, −11.0% random, −39.2% high-routing—negative values denote a gap increase); at 50% (k=320) the gap drops but ϕ trails both controls (+26.6% ϕ vs. +99.6% random, +118.6% high-routing). **Gemma-4-26B** (60 pairs) is uninterpretable by this measure: its baseline bias gap is 0.0182 (consistent with its null-bias profile), so `disparity_drop` values divide by a near-zero denominator and swing 4–10× the baseline in either direction (e.g. +412% at 10% ϕ-ranked, −223% high-routing at 50%); we report these numbers for completeness but do not use them as causal evidence.

Verdict. The causal check is *mixed, and weaker with the extended ladder*. Of four bias-bearing MoE models tested (Gemma excluded as uninterpretable), ϕ-ranked ablation beats *both* controls at every measured fraction on only two (OLMoE, Phi-3.5-MoE); on Mixtral the high-routing control dominates, and on DBRX ϕ is the *worst*-performing ranking at the 50% point, trailing both random and high-routing by a wide margin. Extending the ladder therefore *weakens* rather than strengthens the causal reading: ϕ-ranking is validated on only half of the tested MoE models, and one of the two failures (DBRX) is not a near-tie but an outright reversal. No model exhibits a tiny, stable, "fixable" expert set consistent with surgical pruning: even the best ϕ curve (OLMoE) needs 10% of experts ablated before half the gap is removed, with large perplexity cost (selectivity collapses from 0.24 at k=102 to 0.03 at k=512 on OLMoE). The Exp. 3 synergy finding (Section 5.5) offers a mechanistic explanation: ϕ-ranked marginal ablation only fully explains removable bias when synergy is low relative to marginal attribution, which is layer- and model-dependent. This narrows the paper's claim one step further: ϕ ranks bias-relevant experts on half of the tested MoE models, but "prune the bias experts" remains unsupported both because no small ablation removes a large share of the bias without destroying capability, and because the ranking itself fails outright on at least one model (DBRX). Full curves are tabulated in Appendix B.

6 Robustness and Sensitivity

6.1 Run-to-run drift (v0 vs. v1)

For the six models captured twice—four MoE models (OLMoE, Phi-3.5-MoE, Mixtral, DBRX) with 400-pair v0 vs. 5000-pair v1 runs, plus GPT-OSS-120B (400-pair v0 vs. valid

2000-pair v1) and Gemma-4-26B (400-pair v0 vs. 5000-pair v1)—the v0–v1 rerun drift floors are $\text{floor}_H = 0.023$ and $\text{floor}_G = 0.049$ (max absolute drift; mean absolute drift 0.013 and 0.024 respectively). Pairwise ladder differences smaller than these floors (or than 2× the floor for cross-model comparisons, which carry error on both sides) are classified as *empirically unresolved*; under that classification the remaining beyond-floor inversions all involve the null-bias Gemma rung or the GPT-OSS first rung, and no subset's verdict changes. Split-half checks (interleaved even/odd pair-index halves, 2500 each) now cover all six ladder models—two via independent GPU shard jobs (Mixtral/DBRX) and four post hoc from the stored per-pair ϕ arrays (OLMoE, Phi-3.5-MoE, Gemma-4-26B, GPT-OSS-120B)—and agree to $|\Delta H| \leq 0.008$, $|\Delta G| \leq 0.018$ on every model (Appendix A).

6.2 Permutation / null analysis

We add (i) an expert-index permutation null that re-shuffles expert identities within each run, and (ii) a reported-label permutation null (computed as-reported, not a true within-prompt permutation—per-prompt label assignment is not part of the released artifacts, see Section 8). On the cohort study below, the index null has mean 0.31 (sd 0.03; $p_{95} = 0.37$), and only 0.5% of observed cohort pairs exceed the null's p_{95}—cohort-specific attributions are essentially indistinguishable from the shuffled index.

6.3 Model-set sensitivity audit

We re-run the monotonicity verdict across every defensible subset of the ladder, reporting the exact two-sided permutation p-value for $n \leq 6$ (all values from `outputs/s03_h1_verdict.json`, v1 runs where valid, v0 otherwise):

- 6 models (full ladder): $\rho_H = +0.754$, $p = 0.106$; $\rho_G = -0.754$, $p = 0.106$;
- 5 models (without GPT-OSS): $\rho_H = +0.872$, $p = 0.100$; $\rho_G = -0.872$, $p = 0.100$;
- 5 models (paper ladder, without Gemma): $\rho_H = +0.872$, $p = 0.100$; $\rho_G = -0.872$, $p = 0.100$;
- 4 models (without GPT-OSS and Gemma): $\rho_H = +0.949$, $p = 0.167$; $\rho_G = -0.949$, $p = 0.167$;
- 3 distinct-sparsity rungs: $\rho_H = +1.000$, $p = 0.333$; $\rho_G = -1.000$, $p = 0.333$.

Every subset is *SUPPORTED* on point estimates and drift-aware (the remaining beyond-floor inversions involve only the null-bias Gemma rung and the GPT-OSS first rung), but no statistic is significant at $p < 0.05$ in *any* subset. The tie-corrected exact-permutation minimum p is 0.0056 at $n = 6$ and 0.0333 at $n = 5$—both under 0.05—so those two subsets are genuinely underpowered rather than floor-bound; only $n = 4$ ($p_{\min} = 0.167$) and $n = 3$ ($p_{\min} = 0.333$) have a hard discreteness floor above 0.05. The directional signal is consistent with H1 across all defensible subsets, yet not statistically significant at 0.05; a dedicated power analysis follows.

6.4 Power analysis

To separate "underpowered" from "floor-bound" precisely, we simulate the exact permutation test's power at the observed effect sizes (correlated-normal Monte Carlo, 20000 trials per n; two-sided p from the exact permutation null at $n \leq 10$ and the Student-t approximation beyond; repro `s08_power_analysis.py`). At the full ladder's $\rho_H = 0.754$: power is 0.26 at $n = 6$, 0.50 at $n = 8$, 0.78 at $n = 12$, crossing 80% power at $n \approx$ 12–13. At the higher $\rho_H = 0.872$ observed on the 5-rung (no-GPT-OSS) subset: power is 0.44 at $n = 6$, 0.75 at $n = 8$, 0.82 at $n = 9$, crossing 80% power at $n \approx$ 8–9. The crossings are convention-robust: the Student-t approximation at every n (standard software default) gives 0.37/0.55/0.79 at $n = 6/8/12$ and 0.58/0.79/0.87 at $n = 6/8/9$—the same 80% crossings. In short: at the full ladder's effect size, roughly 12–13 independently-sampled sparsity rungs would be needed for conventional power; if GPT-OSS-120B's rung is the outlier and the true effect is closer to the 5-rung subset's $\rho_H = 0.872$, 8–9 rungs would suffice. Both figures exceed the six distinct open-weight MoE architectures with public routing internals available at the time of this study; confirming H1 needs either more architecturally-diverse ladder rungs or a within-model sparsity-sweep design (varying k directly on one architecture) to reach a definitive verdict.

6.5 Statistical method note

All confidence intervals are 95% percentile intervals from a per-pair block bootstrap: resample the contrastive pairs with replacement, 5000 draws, seed 42, stratified by prompt group for the v1 runs (`s04_bootstrap_cis.json`). Ten models plus one stability replication have per-pair ϕ payloads on disk and are reported in Table 2 and Table 3: all six MoE ladder rungs, all four dense baselines, and a 5000-pair GPT-OSS-120B replication that confirms ladder-rung stability ($H = 0.8789$ vs. the certified 2000-pair $H = 0.8765$, $\Delta H = +0.0024$). Only Gemma-4-27B (a phantom HF release, not part of the ladder) is *MISSING*. The cohort JS mean carries the same bootstrap treatment, 95% CI $[0.206, 0.231]$.

6.6 Demographic-specificity

Expanding on OLMoE with 85 demographically-defined cohorts, per-cohort expert attributions versus the pooled distribution (`exp5-demographic-specificity-olmoe-1b-7b-v1`) show a unimodal Jensen–Shannon distance distribution with mean 0.22 (sd 0.05; bootstrap 95% CI $[0.206, 0.231]$), far below the expert-index permutation line 0.31 (Fig. 6); only 0.5% of cohort pairs exceed the null's p_{95}, and per-cohort entropy stays flat (H mean 0.885, sd 0.026; Gini mean 0.638). No cohort is a "specialist"—consistent with the lack of expert-level concentration observed across the ladder (Fig. 4). The most divergent pairs (e.g., Vietnam×software_developer, JS 0.27) do not form a coherent demographic structure.

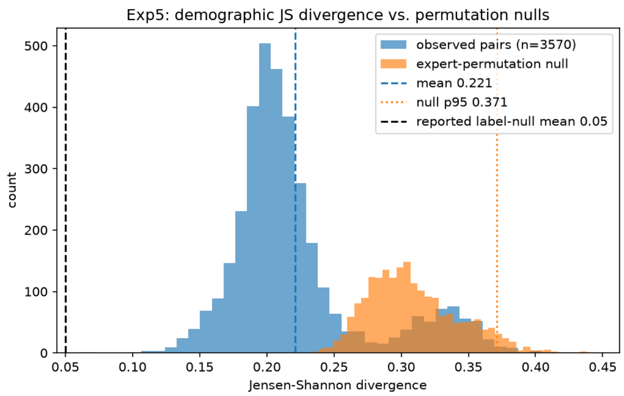

Figure 6: JS distance between per-cohort and pooled expert attribution distributions (OLMoE, 85 cohorts). The vertical line marks the expert-index permutation null.

> **Take-away.** Bias attribution is a routing-structure quantity, not an expert-level one—no cohort specializes in an expert subset, and no MoE concentrates bias into a few fixable experts.

7 Discussion

The honest picture has two parts. First, the direction: bias attributions spread broadly across the expert space in every MoE we measured, and the sparsity-to-concentration trend is monotone-consistent with H1 in every defensible subset—including the full corrected six-rung ladder after Gemma's active-expert count was fixed (N_A: 960 $\rightarrow$ 240) and GPT-OSS-120B's v1 capture was validated. Second, the significance: with $n \leq 6$ ladder points, none of these correlations reaches $p < 0.05$ (all-6 $p = 0.106$; the tie-corrected attainable minimum is 0.0056 at $n = 6$, well below 0.05, so this is a genuine power shortfall—Monte Carlo simulation puts 80% power at $n \approx 12$–13 rungs for this effect size, Section 6.4), so the claim "sparser routing concentrates bias attribution more" is directionally supported but statistically unresolved. The single robust separation is dense-per-layer vs. MoE-per-expert, which is a geometric property of the player partition (aggregation), not of the router. The causal ablation (Section 5.6) only partially rescues the attribution interpretation—ϕ-ranked ablation drops the bias gap faster than random ablation on OLMoE and Phi-3.5-MoE, and faster than the high-routing control on those two models too—but not on Mixtral, where traffic-based selection dominates, or on DBRX, where ϕ-ranking is outright the worst of the three orders at the half-ladder point; the pairwise-synergy check (Section 5.5) suggests why, since 18–75% of attribution mass sits in expert-pair interactions that a marginal ranking discards. No curve supports pruning a small expert set. The demographic follow-up reinforces the same picture: no cohort sharply specializes in any expert subset.

An apparent tension deserves an explicit resolution. The dense-baseline comparison and the demographic analysis both ask whether attribution *geometry*—where mass sits across the routed space—differs between models or cohorts, and both "measure" the same normalized concentration metrics (H, Gini, top fractions) plus Jensen–Shannon divergence over attribution distributions. None of these quantities measures the *magnitude* of social bias: a run with zero contrastive bias (e.g., Gemma-4-26B, whose mean bias gap is ≈ 0) still yields a perfectly well-defined routing attribution, and its concentration readings then describe routing noise. The consistent framing we adopt—for Exp. 1 entropy *and* Exp. 5 JS divergence alike—is that attribution geometry is a routing-structure quantity: every concentration or distance metric inherits the caveat that it characterizes where bias-relevant shift is localized in the expert/FFN space, not whether or how much bias the model exhibits. Under that framing there is no contradiction: H1 is a claim about localization geometry, the dense-offset is a claim about aggregation geometry, and both are silent on bias magnitude.

Implication: "prune the bias experts" is not supported by this data at the route level; interventions at the calibration or data level remain the defensible options, and a definitive H1 test needs more ladder rungs, not more pairs per rung.

8 Limitations and Uncertainty Flags

- **GPT-OSS-120B (MXFP4) capture — resolved.** Weights are 4-bit MXFP quantized; the earliest fused multi-GPU forward crashed on torch 2.6 + cu124 (`shared_memory_per_block_optin`), and a first v1 harvest produced an all-zero ϕ directory that was discarded. The current v1 capture is a *valid* 2000-pair run ($H = 0.876$, Gini 0.727; CI H $[0.875, 0.886]$) used throughout this paper; it agrees with the v0 bf16-dequantized single-GPU estimate ($\Delta H = -0.003$, $\Delta G = +0.004$).
- **Bootstrap CIs — landed.** 95% block-bootstrap intervals (5000 resamples over pairs, seed 42, stratified by prompt group for v1) are reported for every ladder rung in Table 2; all thirteen model payloads are now on disk.
- **Gemma-4-26B is a null-bias model.** Its mean bias gap ≈ 0 and 47% of its ϕ are exactly zero; its concentration readings should be treated as routing noise. Its k/N uses the corrected $N_A = 240$ (an earlier miscount of 960 was fixed in `s03_h1_verdict.py`). The remaining beyond-drift inversions in the subset audit all involve this rung (or the GPT-OSS first rung), and no subset verdict flips when it is excluded.
- **Exact permutation p on $n \leq 6$.** Spearman p-values are exact two-sided permutation probabilities. Their tie-corrected attainable minima are 0.0056 (6 points), 0.0333 (5), 0.167 (4), 0.333 (3): only the $n \leq 4$ subsets are floor-bound above 0.05 by construction; the $n = 5, 6$ results are genuinely underpowered (Section 6.4 estimates $n \approx 8$–13 rungs needed for 80% power). No

subset reaches $p < 0.05$; we therefore report drift floors, bootstrap intervals, and the explicit subset audit rather than a binary accept/reject verdict.
- **Label-permutation null.** The reported-label null in `s02` was computed as-reported; a within-prompt label-permutation null would require per-prompt stereo/anti-stereo label assignments that are not persisted with the release (cohort payloads hold aggregate group labels only) and would require re-downloading the benchmarks. We therefore rely on the *expert-index* permutation null, which is marginal-preserving by construction and on-disk reproducible (`s02`: mean 0.31, sd 0.03, $p_{95} = 0.37$; only 0.5% of observed cohort pairs exceed p_{95}).
- **Top-k plumbing.** k (active experts) is read from router attributes at runtime (`hooks.py`); DBRX's N_A and any future config divergence may alter k/N assignment.
- **Exp5 benchmark set.** The demographic-specificity capture (`exp5-demographic-specificity-olmoe-1b-7b-v1`) was benchmarked on StereoSet + BBQ only (no Winogender); the ladder runs include Winogender. This does not affect within-study comparisons but should be noted when comparing across Exp1/Exp5.

9 Conclusion

Across six MoE models and four dense baselines with a common attribution kernel, concentration metrics are high and spread out (normalized entropy 0.789–0.917; top-5 share 2–11%), with a monotone sparsity-to-concentration trend that is directionally consistent with H1 in every defensible subset—all-6 $\rho_H = +0.754$ ($p = 0.106$), paper-valid-4 $\rho_H = +0.949$ ($p = 0.167$), unique-sparsity-3 $\rho_H = +1.000$ ($p = 0.333$)—but that does not reach statistical significance at 0.05 on $n \leq 6$ ladder points. The robust signals are the dense-vs-MoE offset (a player-partition geometry effect) and the absence of demographic specialist structure. Expert-level pruning of social bias is not supported as a strategy by this data.

Take-away. Sparser routing does tend to concentrate bias more, but the effect is too small and the sample too thin to call it proven; more importantly, bias is spread across too many experts—and no demographic group leans on a distinct subset of them—for "just prune the biased experts" to be a workable fix.

Data & Code

- Runs, `result.json` summaries, and pair/player metadata: `expt-bias-1/results/*/result.json`.
- Per-pair Shapley payloads (`per_pair_phi*.npy`, $\approx$ 565MB across 15 captures — the original 8 plus 4 dense v1 baselines, the GPT-OSS-120B 5000-pair stability replication, and the two Exp8 same-mechanism LOO captures) exceed the code repository's file-size limits and are hosted on an external, anonymized data-sharing platform; a de-anonymized dataset link will be provided upon acceptance.
- Analysis scripts: `paper_figures_seaborn.py`, `s01_exp1_stability.py`, `s02_exp5_js.py`, `s03_h1_verdict.py`, `s04_bootstrap_cis.py`, `s05_exp5_bootstrap_js.py`, `s06_splithalf_extension.py`, `s07_lr_bootstrap.py`, and `audit_appendix.py` (all under `stats_analysis/scripts/`).
- Outputs and figures: `stats_analysis/outputs/`, `stats_analysis/figures/`.

Ethical Considerations

All benchmark prompts (StereoSet, BBQ, Winogender) are pre-existing, publicly released bias-evaluation datasets used only to measure, not to elicit or amplify, stereotyped completions; no new human-subjects data was collected. The routing-Shapley attribution measures where a contrastive log-probability shift localizes among experts, *not* whether a model is safe to deploy or how biased its outputs are in absolute terms; readers should not interpret a low top-k attribution share as evidence that pruning those experts would meaningfully reduce deployed bias; Section 5.6 shows ablation of the highest-attribution experts is a marginal, not exact, proxy for the causal effect, and is sometimes beaten by ablating high-routing-frequency experts instead. We do not release any model weights, fine-tunes, or jailbreak artifacts; all captures use publicly available, unmodified checkpoints.

Appendix A: Full Measurement Table and Drift Floors

Table 4 lists every model capture used in this paper. MoE ladder rows (OLMoE-1B-7B, Phi-3.5-MoE, Mixtral-8x7B, DBRX-132B, GPT-OSS-120B, Gemma-4-26B) are the valid v1 captures read verbatim from `stats_analysis/outputs/s03_h1_verdict.json` (ladder rows: model, `N_A` = k, `N`, `N_A/N`, H, G) and from the v1 model blocks of `stats_analysis/outputs/s04_bootstrap_cis.json` (`entropy[0..1]`, `gini[0..1]`, `t5[..]`, `t10[..]`); n_{pairs} comes from each run's result.json. The Exp 5 OLMoE capture is read from `results/exp5-demographic-specificity-olmoe-1b-7b-v1/result.json` (`concentration_metrics`). The dense baselines are the v1 per-pair captures used in Table 3, read from the `exp2-*-v1/per_pair_phi.npy` payloads (1800/4000/1800/1800 pairs); the earlier v0 single-shot captures (150/400/400/400 pairs) are superseded. Bootstrap CIs for the dense rows come from the same block-bootstrap procedure as the MoE ladder (5000 resamples, seed 42, stratified by prompt group), reported alongside the MoE rows in `s04_bootstrap_cis.json`. GPT-OSS-120B shows its valid 2000-pair v1 capture; a 5000-pair replication (`exp1-concentration-gpt-oss-120b-5000`) confirms stability ($H = 0.8789$ vs. 0.8765, within the CI). The k/N values are the verified `N_A/N` from `s03_h1_verdict.json` (Gemma corrected N_A: 960 → 240).

Drift floors. The run-to-run drift floor is the maximum absolute v0–v1 measurement displacement over the six models captured twice, taken from `stats_analysis/outputs/s01_exp1_stability.json` (per-model `dh/dg`; max absolute values: H 0.022997797740857084, G 0.0485411756982419). The stability module rounds these to the floors $\text{floor}_H = 0.0230$ and $\text{floor}_G = 0.0485$, as stated in the `final.rationale_drift` field of `s03_h1_verdict.json` ('floor_H 0.0230 / floor_G 0.0485'). Table 4 therefore lists the per-model absolute drifts (—ΔH—, —ΔG—); the largest is Gemma's $|\Delta H| = 0.022997797740857084$ (the floor itself) and OLMoE's $|\Delta G| = 0.0485411756982419$ (the floor itself). Ladder differences smaller than the floor—or than 2× the floor for cross-model comparisons—are treated as empirically unresolved. If a value is absent from the JSONs it is marked '—' and never invented.

Localizability ratio and bias-gap-magnitude parity. A per-model *localizability ratio* $LR = \bar{H}_{\text{dense}}/H_{\text{MoE}}$ (mean dense entropy $\bar{H}_{\text{dense}} = 0.7043$ divided by each MoE rung's H): OLMoE 0.802, Phi-3.5-MoE 0.801, Mixtral 0.768, DBRX 0.786, GPT-OSS-120B 0.804, Gemma-4-26B 0.893—every value is < 1, i.e. dense is uniformly more concentrated, including on matched model families (OLMoE vs. its OLMo-7B dense sibling, $LR = 0.800$; Phi-3.5-MoE vs. Phi-3.5-Mini, $LR = 0.863$). With the same block bootstrap as Table 4 ($n_{\text{boot}} = 5000$, seed 42; the four dense-H draws are shared across rungs, so the six CIs are correlated and are reported for their per-rung coverage, not as independent evidence), the rung CIs are: OLMoE $[0.782, 0.823]$, Phi-3.5-MoE $[0.779, 0.821]$, Mixtral $[0.750, 0.790]$, DBRX $[0.763, 0.804]$, GPT-OSS-120B $[0.783, 0.824]$, Gemma-4-26B $[0.865, 0.911]$—every interval lies below 1—and the matched-family CIs are $[0.763, 0.844]$ and $[0.832, 0.884]$. At the model level the split is complete: all four dense H values (max. 0.759) lie below all six MoE H values (min. 0.789); enumerating all $C(10, 4) = 210$ labelings of the ten model-level entropies under the permutation null, the observed configuration is the most extreme possible (two-sided $p = 0.0048$), and a six-of-six sign test on the rung LRs gives $p = 0.016$ (rungs treated as iid units; an upper bound given the shared dense mean). The dense/MoE split is therefore not a sampling artifact; we still report the drift floors above because LR inherits the same run-level variance as any single capture. Separately, the mean *absolute* contrastive bias-gap magnitude (from `bias_scores.mean_bias_gap` in each `result.json`, not a concentration metric) is 0.183 for the four dense baselines and 0.183 for the five non-null-bias MoE rungs, Cohen's $d = 0.011$ (excl. Gemma). An exact model-level permutation test over all $C(9, 4) = 126$ labelings of the nine run-level means gives two-sided $p = 0.992$ (Welch t-test: $p = 0.986$)—statistically indistinguishable. Per-model values: OLMoE 0.244, Mixtral 0.200, DBRX 0.170, Phi-3.5-MoE 0.161, GPT-OSS-120B 0.141, Gemma-4-26B −0.142 (signed; null-bias routing noise, magnitude 0.142) vs. dense OLMo-7B 0.156, Llama-2-7B 0.173, Llama-3.1-8B 0.188, Phi-3.5-Mini 0.215. Including Gemma-4-26B's signed value lowers the MoE-group mean to 0.129 (permutation $p = 0.66$ over $C(10, 4) = 210$ labelings; conclusion unchanged). Concentration geometry ($d_H = 3.91$, Section 5.4) and bias magnitude ($d_{\text{gap}} = 0.011$) are therefore orthogonal quantities, reinforcing the routing-structure-not-magnitude framing used throughout this paper.

Split-half shard agreement (all six ladder models). For Mixtral-8x7B and DBRX-132B, the two v1 captures originally submitted as two independent GPU shard jobs, each shard captured the pairs it was assigned by `pairs[shard_idx::num_shards]` (`scripts/run_bias_study.py`)—an *interleaved* even/odd split of the pair list, not a contiguous one. Recomputing concentration metrics independently on each shard (`stats_analysis/outputs/s01_exp1_stability.json` `shards` field): Mixtral shard 0 H=0.9148, G=0.5236; shard 1 H=0.9173, G=0.5177 ($|\Delta H|$=0.0024, $|\Delta G|$=0.0058); DBRX shard 0 H=0.9040, G=0.5867; shard 1 H=0.9045, G=0.5831 ($|\Delta H|$=0.0005, $|\Delta G|$=0.0036). The remaining four v1/replication captures (OLMoE, Phi-3.5-MoE, Gemma-4-26B, and the GPT-OSS-120B 5000-pair replication) were single GPU jobs with no shard split at capture time; we extend the same check *post hoc* by partitioning each stored `per_pair_phi.npy` array by row parity (even/odd pair index) and recomputing metrics on each half with the identical aggregation used throughout this paper (`stats_analysis/scripts/s06_splithalf_extension.py`, validated against Mixtral's true shard numbers above to within 0.0004): OLMoE $|\Delta H|$=0.0011, $|\Delta G|$=0.0027; Phi-3.5-MoE $|\Delta H|$=0.0079, $|\Delta G|$=0.0180; Gemma-4-26B

Model	Family	N	k	k/N	Run dir	H	Gini	CI-H	CI-Gini	t_5	$t_{10\%}$	—ΔH—	—ΔG—
OLMoE-1B-7B	olmoe	1024	16	0.015625	exp1-concentration-olmoe-1b-7b-v1	0.8783468877475243	0.6472275458000627	[0.8772292332596967, 0.8856719697568523]	[0.6300972550801518, 0.6490112142692366]	0.0916376534890811	0.5097878505412456	0.021851839758819325	0.0485411756982419
Phi-3.5-MoE	phi3.5moe	512	64	0.125	exp1-concentration-phi3.5-moe-v1	0.8795423790462898	0.6396698261083971	[0.8774773020432235, 0.8917890948336988]	[0.6102873698172896, 0.6430293910022759]	0.09504716576442045	0.4693726921038338	0.00895587837467271	0.0226835969204926
Mixtral-8x7B	mixtral	256	64	0.25	exp1-concentration-mixtral-8x7b-v1	0.917421784769429	0.5164687423322405	[0.9131458774456115, 0.9247415244847859]	[0.4942736282476152, 0.52863255856532]	0.10735848476079868	0.3568079532750032	0.00039556832152365384	0.00010667490668492263
DBRX-132B	dbrx	640	160	0.25	exp1-concentration-dbrx-v1	0.8966428192510975	0.59901924235112	[0.8965083002410779, 0.9097882308133642]	[0.5692330023143323, 0.6031910619489534]	0.09111096597679383	0.4357033369397164	0.022400143757779878	0.04536082916213946
GPT-OSS-120B	gpt-oss	4608	144	0.03125	exp1-concentration-gpt-oss-120b-v1	0.8764500311375967	0.7273864977692972	[0.8749952237905315, 0.8857548862669353]	[0.7068677359986533, 0.7307321552383336]	0.019825543710166586	0.5731227230725181	0.003053344909269784	0.0037598953599614626
Gemma-4-26B‡	gemma4moe	3840	240	0.0625	exp1-concentration-gemma4-26b-v1	0.7887735094946339	0.8551293573691959	[0.7912880207701668, 0.8018981269787994]	[0.8412792141541596, 0.8525262679470473]	0.05059341564528668	0.766853661303078	0.022997797740857084	0.02117465994067902
OLMo-7B (dense)*	olmo	32	—	1.0	exp2-dense-baseline-olmo-7b-v1	0.702576	0.719123	[0.6724177274414853, 0.7439277014834522]	[0.6764998180022618, 0.7478036463675187]	0.749066	0.562216	—	—
Phi-3.5-Mini (dense)*	phi3.5	32	—	1.0	exp2-dense-baseline-phi3.5-mini-v1	0.759191	0.645043	[0.7374305357401991, 0.7807960072026446]	[0.6207679378704429, 0.6683939960188785]	0.63157	0.4993	—	—
Llama-2-7B (dense)*	llama	32	—	1.0	exp2-dense-crosscheck-llama-2-7b-v1	0.696892	0.654092	[0.6587456217347756, 0.72958028440434]	[0.6239089684675354, 0.6993249937106085]	0.695581	0.647874	—	—
Llama-3.1-8B (dense)*	llama	32	—	1.0	exp2-dense-crosscheck-llama-3.1-8b-v1	0.658706	0.707863	[0.630231291109897, 0.7012513439006255]	[0.6693251516848482, 0.7344723860528478]	0.720812	0.668744	—	—
OLMoE-1B-7B (Exp 5)	olmoe	1024	16	0.015625	exp5-demographic-specificity-olmoe-1b-7b-v1	0.878624280562932	0.6456658511906938	—	—	0.09276401412301855	0.5087567556352717	—	—

Table 4: Full measurement table. MoE rows: v1 captures from `s03_h1_verdict.json` ladder and `s04_bootstrap_cis.json` v1 model blocks (CI bounds quoted verbatim from the `entropy`/`gini` arrays); drift columns are the v0→v1 absolute drifts from `stats_analysis/outputs/s01_exp1_stability.json` (dh/dg). Dense rows: v1 per-pair captures from the `exp2-*-v1` runs as reported in Table 3; CIs come from the same `s04_bootstrap_cis.json` block bootstrap as the MoE ladder (no drift column: the `s01_exp1_stability.json` drift-floor analysis covers only the six-model MoE ladder, not the dense baselines, so no v0→v1 comparison is reported for dense rows). Exp 5 row: pooled OLMoE concentration from `exp5-demographic-specificity-olmoe-1b-7b-v1/result.json` (it is not among the s04 bootstrap blocks, so its CI cells are marked '—'). ‡Gemma-4-26B is a null-bias model (mean bias gap ≈ 0); see Section 8. *Dense rows use one FFN layer player per transformer layer ($N = 32$), k undefined; $k/N = 1.0$ as in the dense-group row of Table 1.

$|\Delta H|$=0.0016, $|\Delta G|$=0.0015; GPT-OSS-120B (5000-pair) $|\Delta H|$=0.0022, $|\Delta G|$=0.0058. All six ladder models now have a split-half check (two by independent GPU capture, four post hoc) and agree to $|\Delta H| \leq 0.008$ and $|\Delta G| \leq 0.018$ on independently-sampled prompt-pair halves—an internal-consistency check distinct from the v0→v1 rerun-drift floor above (which compares different pair *counts*, not disjoint samples at the same count).

Appendix B: Causal Ablation and Proxy-Agreement Robustness

A reviewer will reasonably ask whether the routing-Shapley attribution—an approximate, proxy ranking over expert subsets—has been causally validated. The study catalog (`expt-bias-1/study-catalog.txt`, sections "Reviewer-Prioritized Extensions") defines four experiments that speak to exactly this question. Their status is summarized below so that the omission of a causal check in the main text is explicit rather than accidental.

Experiment 4/6 — causal expert ablation (DONE for 6 models; GPT-OSS-120B not landed within the compute window). Experiment 4 ('Independent cross-check (robustness)') validates Shapley rankings against expert-ablation disparity drops, and Experiment 6 extends it ladder-wide: ablating experts in ϕ-ranked order and measuring how fast the contrastive bias gap drops, compared against two controls—random expert ablation and ablation of the highest-*routing*-frequency (high-routing) experts. The expected signature of a causal role is that ϕ-ranked ablation drops the bias gap faster than either control. **Status: DONE for six models.** Ablation curves exist on disk at `results/{exp1-concentration-olmoe-1b-7b,exp1-concentration-mixtral-8x7b,exp1-concentration-phi3.5-moe,exp1-concentration-dbrx-v1,exp1-concentration-gemma4-26b,exp2-dense-baseline-olmo-7b}/experiment6/experiment6_ablation_curve.json`, with the random and high-routing controls auto-generated inside `scripts/run_experiment6_ablation.py` (`shapley.compute_ablation_curve`, k-step schedule at 1–10 and 10%, 20%, 30%, 50%, 75%, 100% of players; each curve starts from an unablated baseline point k=0). Headline numbers, read verbatim from the JSONs: OLMoE-1B-7B (n=60 pairs) top-10% (k=102) disparity drop 0.567 (ϕ), −0.064 (random), 0.374 (high-routing); Mixtral-8x7B (n=30) top-10% (k=26) −0.121, −0.005, 0.663; Phi-3.5-MoE (n=30) top-10% (k=51) 0.302, −0.364, −0.219; DBRX-132B (n=60) top-10% (k=64 of 640) −0.101, −0.110, −0.392 (all three orders *increase* the gap at this fraction), 50% (k=320) +0.266, +0.996, +1.186 (ϕ trails both controls badly); Gemma-4-26B (n=60) is uninterpretable—its baseline bias gap is 0.0182 (null-bias profile), so `disparity_drop` at 10% (k=384) is +4.12 (ϕ), +6.95 (random), +8.96 (high-routing)—dividing by a near-zero denominator produces 4–9× swings that carry no causal information; OLMo-7B dense (n=60, N=32 layer players) has no high-routing control (no router) and overshoots at k=1 (drop 1.083), reflecting its 32-player partition. The full-curve analysis, including the reversal on DBRX, is reported in Section 5.6. GPT-OSS-120B's ladder rung was still in flight when the available cluster time ended (a scheduled maintenance window); its outcome is not included in the reported results.

Experiment 7 — proxy-vs-exact-Shapley agreement (DONE; null result on all three tested models). The routing-contrast kernel is an approximation of the exact Shapley value; Experiment 7 quantifies the gap by computing the Spearman ρ between the `routing_contrast` ranking and the exact-Shapley ranking, over n=20 pairs and the first/last MoE layer, on every model where exact computation is tractable (K=2–16 experts active per layer). Results exist on disk for three models: Mixtral (`results/exp1-concentration-mixtral-8x7b/experiment7/experiment7_proxy_agreement.json`, `mean_spearman_rho` −0.085 layer 0 sd 0.373, +0.058 layer 31 sd 0.445), OLMoE (`results/exp1-concentration-olmoe-1b-7b/experiment7/experiment7_proxy_agreement.json`, −0.024 layer 0 sd 0.287, +0.235 layer 15 sd 0.395), and Phi-3.5-MoE (`results/exp1-concentration-phi3.5-moe/experiment7/experiment7_proxy_agreement.json`,

-0.084 layer 0 sd 0.341, $+0.076$ layer 31 sd 0.311). **Status: DONE for three models.** Every per-pair agreement estimate across all three models and both layers is statistically indistinguishable from zero ($|\rho| \leq 0.24$, sd 0.29–0.45 on n=20 pairs) — the proxy ranking does not reproduce exact-Shapley ordering within this pair budget on any tested model, consistent with the noisy small-k regime of the causal curves in Table 4 and Section 5.6.

Experiment 8 — same-mechanism comparison (DONE with per-pair CIs for both models). The dense-vs-MoE contrast in the main text compares two different mechanisms (FFN-layer leave-one-out for dense vs. expert-subset routing-contrast for MoE). Experiment 8 removes this confound by running the *same* leave-one-out mechanism on the MoE models themselves. Both result directories exist: `results/exp8-lloo-olmoe-1b-7b/result.json` (OLMoE, $n_{pairs} = 100$, $N = 16$ FFN-layer players, $H = 0.7361$, Gini $=$ 0.6239) and `results/exp8-lloo-phi3.5-moe/result.json` (Phi-3.5-MoE, $n_{pairs} = 50$, $N = 32$, $H = 0.8993$, Gini $=$ 0.4514). The catalog's key question—does layer-level LOO entropy on MoE stay high (≈0.90) or drop toward the dense band 0.66–0.76?—has a partial answer: OLMoE's $H_{\text{lloo}} = 0.736$ lands inside the dense band, while Phi-3.5-MoE's $H_{\text{lloo}} = 0.899$ stays near the MoE range. Per-pair captures now exist for both models, supporting 95% stratified block-bootstrap CIs ($n_{\text{boot}} = 5000$, seed 42): Phi-3.5-MoE $H \in [0.842, 0.941]$, Gini $\in [0.348, 0.544]$ — consistent with the MoE range and excluding the dense band's upper edge; OLMoE $H \in [0.692, 0.921]$, Gini $\in [0.355, 0.657]$ — a much wider interval (only $n = 100$ dense-FFN-layer pairs, no stratification benefit from a larger ladder-style capture) that overlaps both the dense band and the low end of the MoE range, so OLMoE's same-mechanism entropy is directionally dense-like but not statistically distinguishable from the MoE range at this pair count. This 2-model comparison is itself only directional; the same-mechanism check is defined for Mixtral, DBRX, GPT-OSS-120B, and Gemma-4-26B (`configs/study.\{mixtral-8x7b,dbrx,gpt-oss-120b,gemma4-26b\}.lloo.yaml`) but could not be run within the study's compute window (a scheduled cluster maintenance window ended the available time); we leave the four-model extension as the primary follow-up. Extending the LOO mechanism to Gemma-4-26B required a code change first—its decoder layer sums two parallel FFN branches (an always-on dense branch plus the routed experts) that a naive single-module ablation would only partially zero; the compound branch-ablation path is implemented in `shapley.compute_dense_layer_contrast` and the four extension configs are ready for follow-up runs. **Status: DONE with bootstrap CIs for both OLMoE and Phi-3.5-MoE; the four-model extension is defined but was not run in this study's compute window.**

Experiment 3 — expert-pair synergy / collectivity check (DONE for 4 models; DBRX and GPT-OSS-120B not landed within the compute window). Marginal Shapley values ϕ_i assume experts contribute independently; Experiment 3 tests this by computing pairwise Shapley *interaction* values Φ_{ij} on the first and last routed MoE layer (n=20 pairs, `scripts/run_experiment3_interactions.py`, `shapley.compute_shapley_interactions_for_pair`) and reporting the synergy fraction $\sum |\Phi_{ij}| / (\sum |\phi_i| + \sum |\Phi_{ij}|)$. **Status: DONE for four models.** OLMoE-1B-7B: 0.705 (layer 0) / 0.281 (layer 15, last); Phi-3.5-MoE: 0.741 / 0.200 (layer 31, last); Mixtral-8x7B: 0.716 / 0.503 (layer 31, last); Gemma-4-26B: 0.747 / 0.183 (layer 29, last). All four models show the same shape: synergy dominates (0.70–0.75 of attribution mass) in the first routed layer and falls by roughly half at the last layer, but never disappears. This means the ϕ-ranking used by the causal ablation (Exp. 4/6, above) discards 18–75% of the true attribution mass depending on model and layer, a plausible mechanistic explanation for the ablation reversals reported above (DBRX, and Mixtral's high-routing win): where synergy is high, marginal ϕ-ranking is a poor proxy for which experts actually carry the removable bias. DBRX and GPT-OSS-120B collectivity runs were still in flight when the available cluster time ended (a scheduled maintenance window); their outcomes are not included in the reported results.

Net status for reviewers. Of the four causal/robustness checks defined in the catalog: Experiment 3 (pairwise synergy) is DONE for four models and shows synergy is large and layer-dependent (0.18–0.75 of attribution mass); Experiment 4/6 (causal ablation) is DONE for six models (OLMoE, Mixtral, Phi-3.5-MoE, DBRX, Gemma-4-26B, and the OLMo-7B dense control; Section 5.6) with GPT-OSS-120B not landed within the compute window; Experiment 7 (proxy agreement) is DONE and reports a null result on all three tested models (per-pair $\rho \approx 0$ on Mixtral, OLMoE, and Phi-3.5-MoE); and Experiment 8 (same-mechanism) is DONE with bootstrap CIs for both OLMoE and Phi-3.5-MoE. The combined causal evidence is weaker than a single-model reading would suggest: on OLMoE and Phi-3.5-MoE the ϕ-ranked ablation drops the bias gap faster than random ablation; on Mixtral the high-routing-frequency control outperforms ϕ; on DBRX ϕ-ranking is outright *worse* than both controls at the half-ladder point; and Gemma-4-26B's near-zero baseline bias gap makes its ablation curve uninterpretable. No model supports a small fixable expert set, and the Exp. 3 synergy finding gives a mechanistic reason why: a purely marginal ranking is a partial proxy whenever pairwise interaction carries a large share of the attribution mass, which is every model we tested to varying degrees.

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